

Murali Karthik Ganji

(520) 437-1149 | muralikarthikganji@gmail.com | [linkedin.com/in/murali-karthik-ganji](https://www.linkedin.com/in/murali-karthik-ganji) | murali-karthik-ganji.vercel.app

Computer Science student at the University of Arizona with experience in software development, applied AI research, machine learning, and data engineering. Experienced in building software applications, developing Python-based data pipelines, preparing and analyzing research datasets, and applying machine learning and computer vision techniques to real-world problems. Strong foundation in software engineering, data-driven problem solving, and collaborative technical development.

EDUCATION

University of Arizona, College of Science

Bachelors in Computer Science

- **Minor:** Statistics & Data Science
- **GPA:** 3.10
- **Honors:** Global Wildcat Scholarship, Dean's List Full Time Award – Spring 2024

Tucson, Arizona

Expected Graduation Date: December 2026

Relevant Coursework: Machine Learning, Natural Language Processing, Data Engineering, Systems Programming, Software Development, Computer Organization, Statistical Computing, Web Programming, Analysis of Discrete Structures, Comparative Programming Languages, Linear Algebra, Vector Calculus

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, JavaScript, C, R, Swift, Haskell, Prolog, Ruby

Frameworks & Libraries: React, Node.js, Express.js, Pandas, NumPy, scikit-learn, LightGBM

AI, Computer Vision & Data Engineering: OpenCV, FFmpeg, Supervisely, Image Processing, Semantic Segmentation, Data Pipelines, ETL, Feature Engineering, Data Cleaning, Statistical Modeling, Time Series Forecasting

Databases: MongoDB, MySQL, PostgreSQL

Cloud & Developer Tools: AWS, Docker, Git, GitHub, Bash, Linux/UNIX, VS Code

Software Engineering: Data Structures, Algorithms, OOP, Design Patterns, RESTful APIs, MVC Architecture, System Design, SDLC, Agile, Version Control, Testing, Debugging

Office & Productivity: Microsoft Excel, Word, PowerPoint, Outlook, Google Workspace

Operating Systems: Linux/UNIX, Windows, macOS

PROFESSIONAL & RESEARCH EXPERIENCE

Undergraduate Research Assistant

University of Arizona – Department of Electrical & Computer Engineering

Retinal Surgical Video Analysis & Computer Vision

Tucson, AZ

July 2026 – Present

- Develop Python-based preprocessing pipelines for retinal surgical video analysis and computer vision dataset preparation.
- Implement blur detection, SSIM-based filtering, metadata generation, and automated quality-control workflows to improve research dataset quality.
- Prepare annotated semantic-segmentation datasets and case-level data splits to support machine learning model development and evaluation.

VSI-VERIFY – Multi-View Visual Dataset Research

- Support the development of Dataset v1 by designing structured multi-view and multi-state image collection workflows, metadata schemas, and quality-control procedures.
- Develop reproducible capture and data-organization processes, including standardized file structures, naming conventions, documentation, and dataset validation.
- Collaborate with researchers to establish collection protocols and prepare high-quality visual data for future AI and computer vision research.

Web Development Intern

Susampada Developers

Hyderabad, India

May 2023 – July 2023

- Maintained and updated website content while improving usability and responsive performance.
- Assisted with SEO optimization, digital content management, and website maintenance.
- Coordinated with clients to gather project requirements, maintain documentation, and support web development initiatives.

PROJECTS

Forecasting Web Traffic for 145,000+ Wikipedia Pages (CSC 480 – Machine Learning Final Project) | [GitHub](#)

- Engineered scalable Python data pipelines to process and analyze traffic data for over **145,000 Wikipedia pages**, supporting feature engineering, model training, and evaluation.
- Designed lag-based and rolling-window features to capture temporal patterns and improve forecasting performance on sequential datasets.
- Implemented a hybrid **LightGBM-LSTM** forecasting model that improved **SMAPE accuracy by approximately 7%**.
- Conducted large-scale preprocessing, model validation, and reproducible experimentation for time-series forecasting research.

Student Performance Prediction System (AI / Machine Learning Project) | [GitHub](#)

- Built machine learning models to predict student academic performance using behavioral and academic datasets.
- Implemented **Decision Tree** and **Random Forest** algorithms for classification and regression tasks.
- Evaluated model performance using **MAE** and **R²** metrics to assess prediction accuracy and generalization.
- Analyzed feature importance to identify variables most strongly associated with academic outcomes.

Research Data Engineering Project | [GitHub](#)

- Designed an end-to-end data pipeline to collect, clean, transform, and analyze datasets related to social media usage and student well-being.
- Automated ETL workflows using Python to improve data preparation and analytical efficiency.
- Performed exploratory data analysis and generated insights through data transformation and visualization techniques.

CineTicket – Movie Ticket Reservation System | [GitHub](#)

- Built a full-stack movie ticket reservation platform using **Node.js**, **Express.js**, **MongoDB**, and **RESTful APIs**.
- Implemented secure user authentication, session management, booking history, and database integration using **Express-session** and **Mongoose**.
- Developed a dynamic frontend using the **Fetch API** to support asynchronous movie browsing and ticket reservation workflows.
- Utilized Git/GitHub for version control, collaborative development, and project management.

fintrackr – Financial Analytics & Statistical Computing Toolkit (R) | [GitHub](#)

- Built an R-based financial analytics toolkit supporting budgeting, investment analysis, and debt evaluation.
- Implemented reusable analytical functions using **tidyverse** and **rlang** for financial summarization and comparative analysis.
- Created interactive visualizations with **ggplot2** to identify financial trends and support exploratory analysis.
- Applied statistical computing principles to develop maintainable and reproducible analytical workflows.

Restaurant Management System – Collaborative Software Engineering Project (CSC 335) | [GitHub](#)

- Collaborated in a multi-developer GitHub repository using **Agile**, pull requests, code reviews, version control, and testing workflows.
- Designed application components following the **Model–View–Controller (MVC)** architecture to improve maintainability and separation of concerns.
- Managed feature integration, branch management, debugging, and collaborative software development using Git.

FinTrack iOS Application – Personal Finance Management App (Swift) | [GitHub](#)

- Developed a personal finance management application using **Swift** and **SwiftUI** for expense tracking, budgeting, and financial reporting.
- Designed responsive user interfaces and implemented local data storage for transaction and category management.
- Applied object-oriented design principles to create maintainable and scalable application components.
- Integrated charts and interactive visualizations to improve usability and user experience.

CERTIFICATIONS

- Machine Learning Specialization – Stanford University & DeepLearning.AI (2026)
- Advanced Learning Algorithms – Stanford University & DeepLearning.AI (2026)
- Supervised Machine Learning: Regression & Classification – Stanford University & DeepLearning.AI (2026)
- AWS Educate – Introduction to Generative AI Badge (2026)
- Artificial Intelligence Fundamentals – Great Learning (2025)
- Front-End Development (HTML) – Great Learning (2024)
- Infosys Catch Them Young (2021)

LANGUAGES, SKILLS, ACTIVITIES & INTERESTS

Languages: English, Telugu, Hindi and French

Soft Skills: Leadership, Organization, Attention to Detail, Communication, Team Collaboration, Time Management, Problem Solving

Activities: Student Council, TECHNIX-2019, Youth Parliament-2019, Infosys Catch Them Young-2021

Interests: Software Engineering, Distributed Systems, Cloud Computing, Machine Learning, Open-Source Software, Scientific Computing